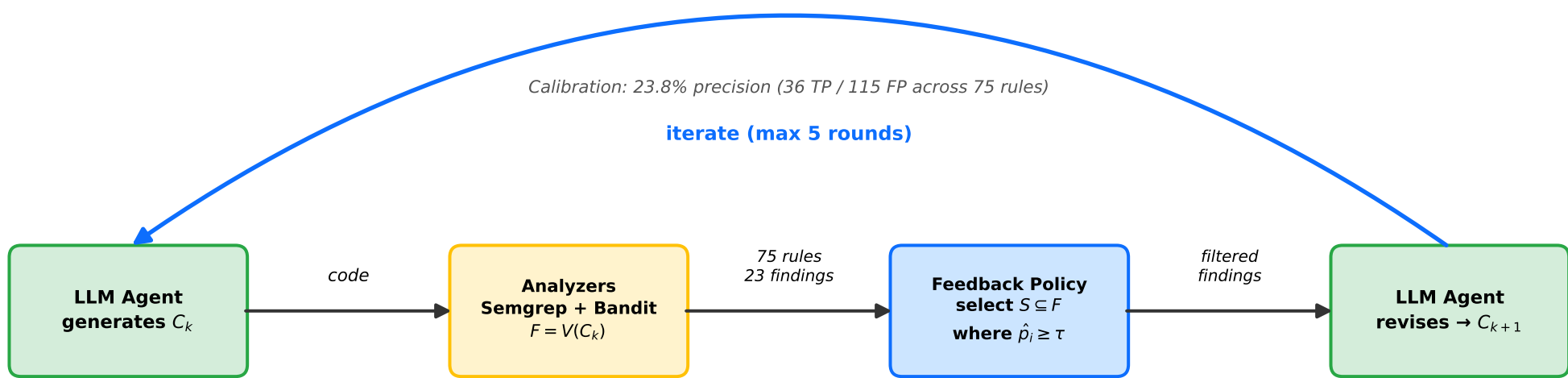




Per-finding decision: surface finding i only if $\hat{p}_i \geq \tau^* = r/(q+r)$

True Positive surfaced
prob p_i : finding IS a vulnerability
LLM fixes with prob q
Haiku: $q=0.26$ | Qwen: $q=0.42$

→ JointPass improves

False Positive surfaced
prob $1 - p_i$: finding is NOT a vulnerability
LLM "fix" causes regression with prob r
Haiku: $r=0.00$ | Qwen: $r=0.43$

→ JointPass degrades

Optimal threshold: $\tau_{\text{Haiku}}^* = 0/(0.26 + 0) = 0$ (surface everything) $\tau_{\text{Qwen}}^* = 0.43/(0.42 + 0.43) = 0.50$ (filter low-precision rules)